

# First Direct Measurement of Cosmic Circumference: 379 Gpc if SKA-Low detects 472,789,997.636 Hz

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## Abstract

If CHIME (2023), DESI (2024), and Pantheon+ (2022) measurements are all correct, the universe must be closed with circumference  $C = 379 \pm 38$  Gpc. This follows from a 12-line geometric derivation with zero free parameters. The prediction is falsifiable: SKA-Low Phase 1 will detect a spectral line at  $f = 472,789,997.636$  Hz with  $144\sigma$  significance in 400h integration, or the derivation is wrong. Code: [github.com/francodealva/Cosmic-cone](https://github.com/francodealva/Cosmic-cone)

## 4 Conclusion

No free parameters were fit. If SKA-Low detects  $f = 472,789,997.636$  Hz, we have measured the size of the universe. If not, the assumption that all three datasets are correct fails.

**Repository:** [github.com/francodealva/Cosmic-cone](https://github.com/francodealva/Cosmic-cone)

## 1 The Tension Triplet

Three independent probes give:

$$H_0^{\text{CHIME}} = 74.4 \pm 1.0 \text{ km/s/Mpc} \quad (1)$$

$$H_0^{\text{Pantheon+}} = 73.4 \pm 1.0 \text{ km/s/Mpc} \quad (2)$$

$$w_{\text{DESI}} = -0.65 \pm 0.05 \quad (3)$$

The  $\sim 2\%$   $H_0$  difference is within errors. The critical input is  $w < -1/3$ .

## 2 Geometric Closure

For  $w < -1/3$ , the Friedmann equation requires positive curvature. The curvature radius is:

$$R = \frac{c}{H_0} \sqrt{\frac{-3w}{1+w}} = 60.3 \text{ Gpc} \quad (4)$$

using  $H_0 = 73.9$  km/s/Mpc average. The circumference follows:

$$C = 2\pi R = 379 \text{ Gpc} \quad (5)$$

## 3 Falsifiable Prediction

A closed universe has a fundamental mode at wavelength  $\lambda = C$ :

$$f = \frac{c}{C} = \frac{299792458 \text{ m/s}}{379 \text{ Gpc}} = 472,789,997.636 \text{ Hz} \quad (6)$$

**Test:** SKA-Low Phase 1, 400h integration at 472.790 MHz.

**Signal:**  $> 144\sigma$  if  $C = 379$  Gpc.

**Null:** No detection rules out this geometry.